

Adarsh Agarwala — Profile Knowledge Base

AI/ML & Quant - IIIT Pune - Pune, India

Who is Adarsh

Adarsh Agarwala is a Computer Science undergraduate (B.Tech CSE, 2025-2029) at IIIT Pune, working at the intersection of AI/ML and quantitative finance. He builds production-grade ML systems, CUDA tensor engines, RAG pipelines, and low-latency C++ - all on a strong mathematics foundation in linear algebra, probability, and stochastic calculus. He is based in Pune, India (Asia/Kolkata).

Focus and interests

AI / ML: retrieval-augmented generation, hallucination detection, LoRA fine-tuning, and AI agents.

Quant: statistical arbitrage, options pricing, factor models, and low-latency trading systems.

Systems: modern C++ (17/20), CUDA, and performance engineering.

Adarsh is open to AI/ML and quant internships, freelance, and contract work.

Tech stack

Python and PyTorch for ML; C++17/20 and CUDA for systems and quant; NumPy, Pandas, statsmodels, and scikit-learn for data; React, Node, Express, and MongoDB on the web side.

Project: Evidence-Aware Medical RAG (AI/ML)

A production-grade medical RAG pipeline doing both dense and vectorless retrieval, then verifying every generated claim against its sources with an NLI-based hallucination detector and LoRA fine-tuning. GitHub: <https://github.com/agarwaladarshcoding-maker>

Project: Mean-Reverting Pairs Trading (Quant)

A full statistical-arbitrage pipeline on GOOGL/MSFT - cointegration testing, rolling hedge ratios, and z-score signals over three years of real market data. GitHub: <https://github.com/agarwaladarshcoding-maker>

Project: HFT Order Matching Engine (Systems)

A low-latency limit order book in modern C++ with price-time priority matching, O(1) lookups, and a cache-optimised memory layout. GitHub: <https://github.com/agarwaladarshcoding-maker>

Project: AgentWatch (AI/ML)

A macOS monitor that intercepts CLI AI tools in real time and classifies their activity with a custom ML model at about 99% accuracy. GitHub: <https://github.com/agarwaladarshcoding-maker>

Project: Monte Carlo Option Pricer (Quant)

Prices down-and-out barrier options over 100k simulated geometric-Brownian-motion paths in under 1.8s, with early-exit optimisation. GitHub: <https://github.com/agarwaladarshcoding-maker>

Project: PCA Factor Model (Quant)

A from-scratch PCA factor model using SVD in NumPy, with no ML libraries, preserving 95%+ of variance across multi-asset returns. GitHub: <https://github.com/agarwaladarshcoding-maker>

Experience: Smart India Hackathon

Mentored and contributed at Smart India Hackathon, helping teams turn ideas into working prototypes under tight deadlines.

Experience: India Innovates 2025

Led Team SHA256 at India Innovates 2025, owning architecture and coordination for the build.

Experience: IIT Pune Coding Club

Active member of the IIT Pune Coding Club, writing the daily 'Day X of Infinity' series on C++ internals and quant finance.

Achievements

JEE Main 2025 All India Rank 17,517 (99th percentile, 97th percentile in Maths). Top 5% at IIT Pune with a CGPA of 9.38. First place in the First-Year Competitive Programming Competition. Team Lead at India Innovates 2025. Author of the 'Day X of Infinity' writing series.

Education

B.Tech in Computer Science and Engineering at IIT Pune, 2025-2029. Current CGPA 9.38 (top 5% of the cohort). JEE Main 2025 All India Rank 17,517 - 99th percentile overall and 97th percentile in Mathematics. Strong theoretical base in linear algebra, probability, statistics, and stochastic calculus that underpins both his ML and quant work.

What Adarsh is looking for

Adarsh is actively open to internships, freelance, and contract work in AI/ML and quantitative finance. He is most excited by applied machine learning, LLM and RAG systems, quantitative research and trading, and low-latency systems engineering. He is comfortable working remotely across time zones from Pune, India (IST, Asia/Kolkata).

How Adarsh works

Adarsh builds end to end and from first principles - for example implementing PCA from scratch with SVD instead of calling a library. He cares about correctness and verification (his medical RAG checks every claim against its sources) and about performance (cache-aware C++, early-exit Monte Carlo). He documents and teaches what he learns through his daily 'Day X of Infinity' writing series.

Guiding ideas

A few principles Adarsh lives by: 'I figure things out.' 'Jack of all trades, master of none, but often better than most.' 'I only see my goals.' And: 'AI and quant are really just maths, and I love maths.'

Skills in detail

Languages: Python, C++ (17/20), and some JavaScript/TypeScript. ML: PyTorch, scikit-learn, RAG pipelines, LoRA fine-tuning, and NLI-based hallucination detection. Quant: NumPy, Pandas, statsmodels, cointegration and statistical arbitrage, options pricing, Monte Carlo methods, and PCA factor models. Systems: CUDA, low-latency data structures, and performance engineering. Web: React, Node, Express, and MongoDB. Competitive programming on Codeforces (handle AdarshAg).

Contact and links

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